BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Linear Search and Algorithm Tracing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Identify the flowchart shape used for a decision.

2. What does iteration mean?

3. Write a Boolean condition to check if x is greater than 10.

4. Trace: x = 4; if x > 3 then x = x + 5 endif; print(x).

Today's objectives

Describe how linear search checks each item in turn. Trace a small search using indices.

Key vocabulary - Read, cover, recall

linear search
Checks items one by one for a target.

target
Value being searched for.

index
Position used to access an array item, starting at zero.

comparison
Checking an item against a value or condition.

Write one definition in your own words:

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Learn and practise

New learning - 2.1.3

A sequential scan	Linear search starts at one end of a collection and compares one element at a time with the target.
Zero-based indices	The first array element is at index 0, then 1, 2, etc. Do not confuse index with value.
Stopping	If the target is found, report its index. If the list ends without a match, report not found.
One strength	Linear search works even if values are unsorted, though it may need to inspect the entire list.

Linear search for target 8: unsorted data is fine

Index	0	1	2	3	4
Value	3	9	1	8	5

3 -> 9 -> 1 -> 8: found at index 3 (four checks).

I Do - Worked example

Trace linear search for 8 in [3, 9, 1, 8, 5]. Give the comparisons and index.

1. Start at index 0 and compare 3 with 8: no match.
2. Index 1: 9 is not 8.
3. Index 2: 1 is not 8.
4. Index 3: 8 matches the target.
5. Report found at index 3 after four comparisons.

We Do - Guided practice

Search [7, 2, 6, 4] for 6. Which elements are compared?

Use the model. Write your answer in BLACK PEN:

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Think and write

Code First - Predict > Write > Test > Improve

Read the code before you write or test anything.

```
array registers = ["MAR","MDR","PC"]
for i = 0 to 2
  if registers[i] == "PC" then
    print(i)
  endif
next i
```

Trace a search for a previously learned CPU register name.

1. Predict the index printed.

2. Change the target to "MDR".

3. State what the PC and MDR hold in a real CPU.

You Do - Independent exam questions

Q1. State whether linear search requires a sorted list. [1 mark]

Q2. Search [9, 4, 1, 6] for 1. State each comparison and the zero-based index returned. [3 marks]

Q3. Explain how linear search would find 7 in an unsorted list [3, 8, 7, 1], or report failure. [4 marks]

Exit Ticket - True or False?

Answer each statement before turning to the answers.

1. Linear search can only work when a list is already sorted. ☐ T ☐ F

2. Index 0 refers to the first item of an array. ☐ T ☐ F

3. A linear search can inspect unsorted data. ☐ T ☐ F

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Review & improve

GREEN PEN: Tick what is correct, then use the model to fix mistakes.
Do not just copy answers; explain what you changed.

Do Now - Answers

Q1 Diamond. ☐

Q2 Repeating a set of instructions. ☐

Q3 x > 10. ☐

Q4 9. Condition true so 4 + 5 = 9. ☐

Code First - Model

PC is at zero-based index 2; MDR is at index 1. PC holds the address of the next instruction. MDR holds data or instructions being transferred to or from memory. The loop is a simplified linear search.

We Do - Model answer

7, 2, then 6. Found at index 2 after three comparisons.

Exit Ticket - Check

1. FALSE - False: revise the new learning.
2. TRUE - True: the source uses zero-based indices.
3. TRUE - True: linear search checks items sequentially.

You Do - Self-mark your exam answers

Q1 No. Linear search can examine items in an unsorted list [1]. ☐

Q2 Compare 9 [1], 4 [1], then 1 at index 2 [1]. ☐

Q3 Start at first element [1]; compare sequentially 3, 8, 7 [1]; report index 2 when found [1]; if end reached with no match report not found [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW complete answer. Show working: